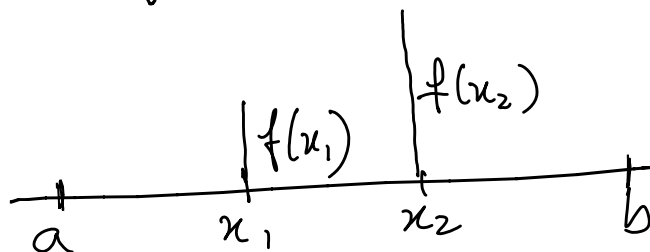


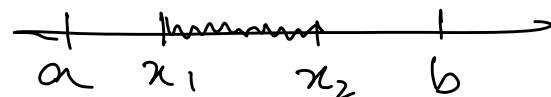
We say a function f is increasing on an interval I if for any $x_1, x_2 \in I$ such that $x_2 > x_1$, then $f(x_2) > f(x_1)$.



Theorem:

If $f' > 0$ on an interval then f is increasing on that interval.

Proof: Let the interval be $[a, b]$. Pick any two points x_1 and x_2 in $[a, b]$ such that $x_2 > x_1$.



1. f cont on $[x_1, x_2]$
2. f db on (x_1, x_2)

So a $c \in (x_1, x_2)$ exists such that

$$\underbrace{f'(c)}_{+} = \frac{f(x_2) - f(x_1)}{\underbrace{x_2 - x_1}_{+}} \text{ Must be pos.}$$

$$\Rightarrow f(x_2) - f(x_1) > 0$$

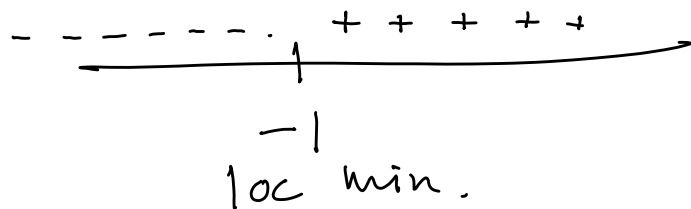
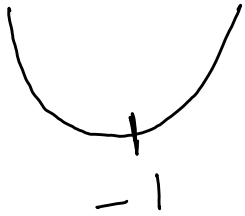
$$\Rightarrow f(x_2) > f(x_1)$$

Example

Find the interval on which $f(x) = xe^x$ increases

$$f'(x) = e^x + xe^x = e^x(1+x).$$

$$f' > 0 \Rightarrow 1+x > 0 \Rightarrow x > -1.$$



First derivative test. (Most useful).

1. If f' changes from positive to negative then f has a local maximum in the point.
2. If f' changes from negative to positive then f has a local minimum in the point.

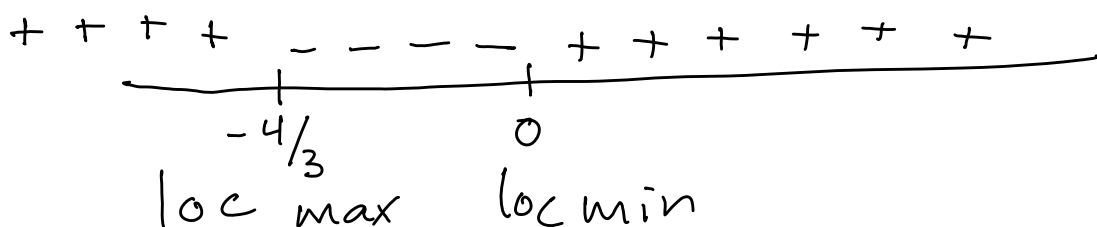
Example

$$f(x) = x^3 + 2x^2$$

$$\Rightarrow \boxed{f'(x) = 3x^2 + 4x} = 0$$

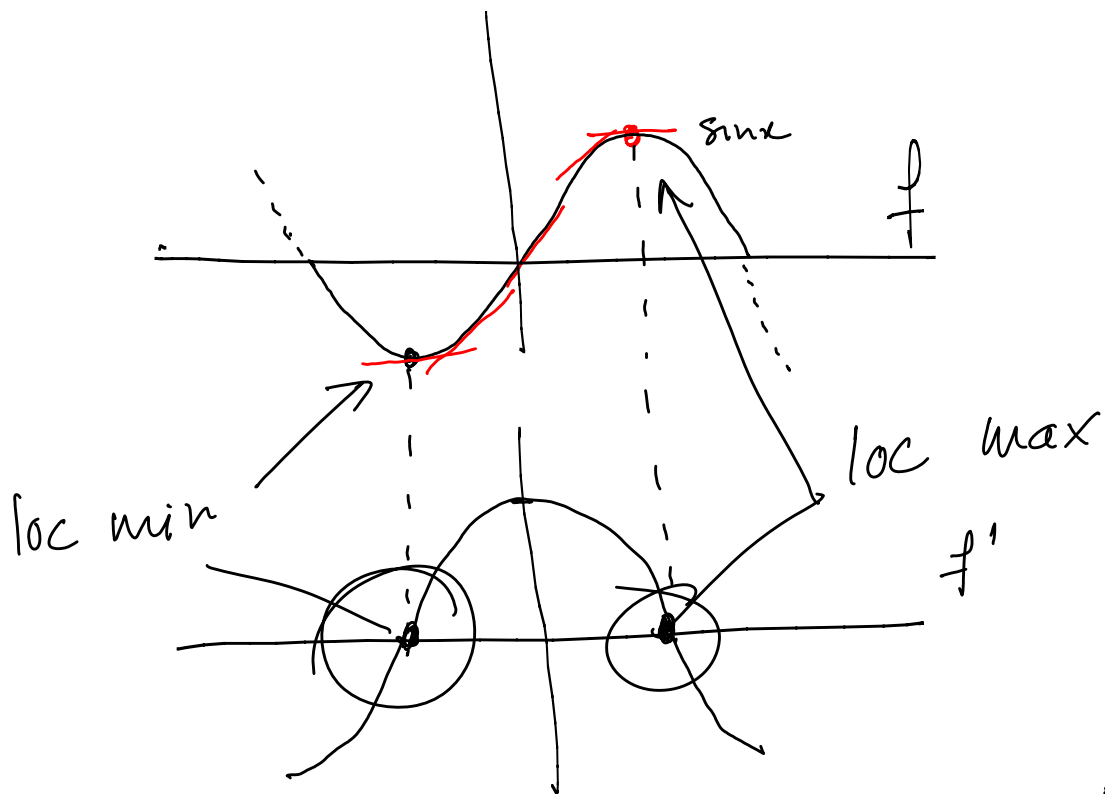
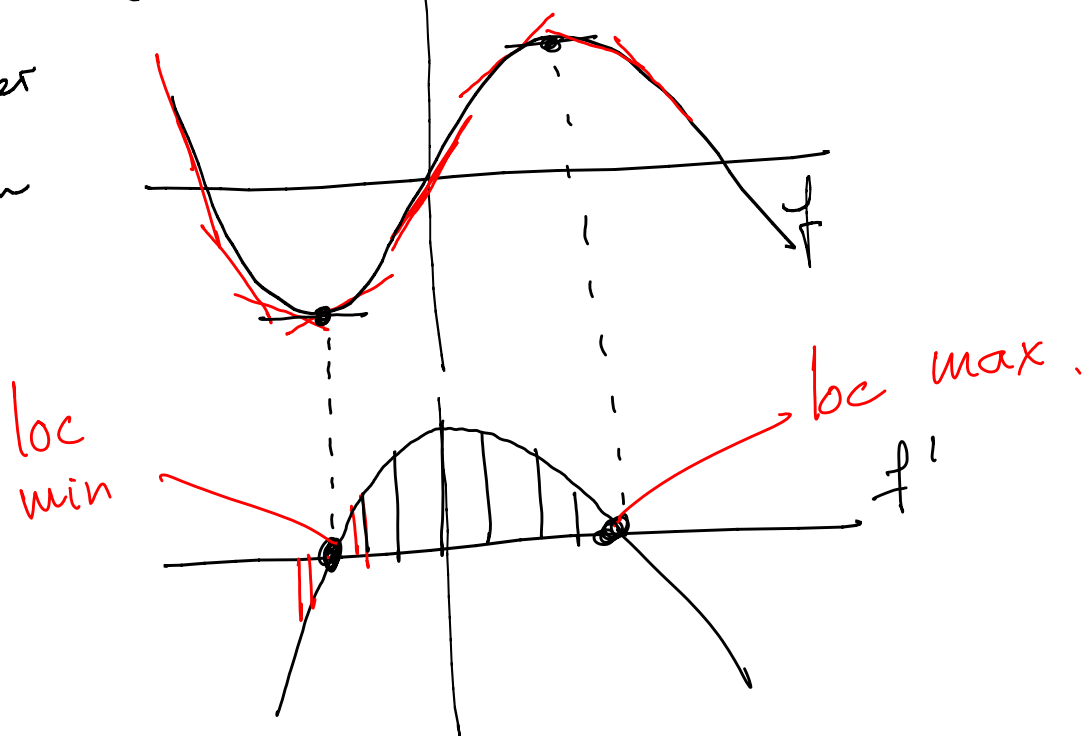
$$\Rightarrow x(3x+4) = 0$$

$$\Rightarrow x = 0 \text{ or } x = -4/3.$$



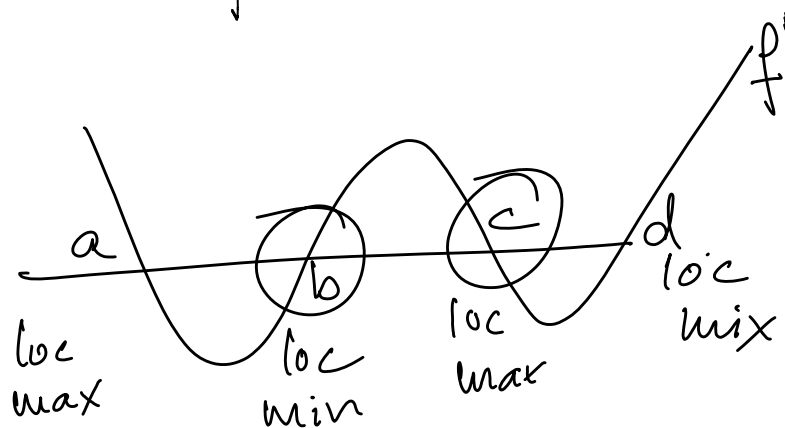
Graphically: From f to f'

Consider
the
function



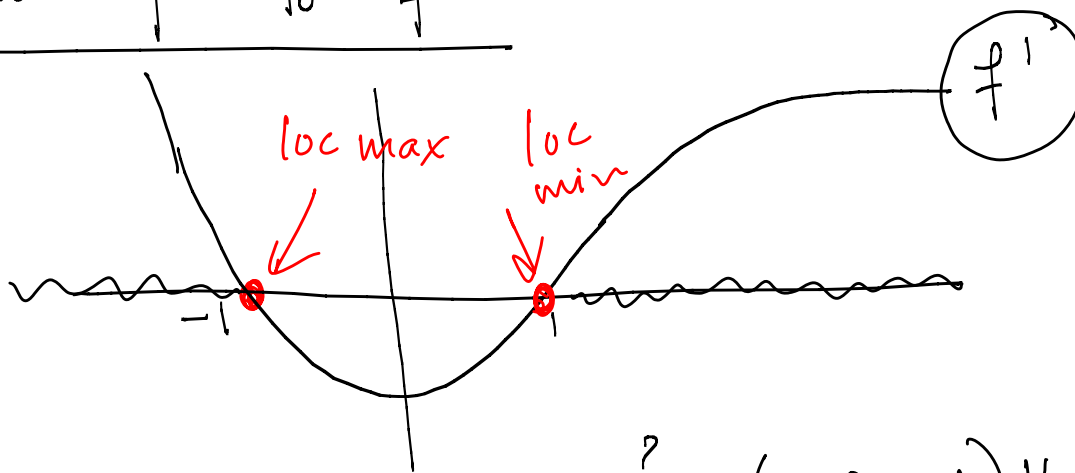
One more:

f ?



From f' to f

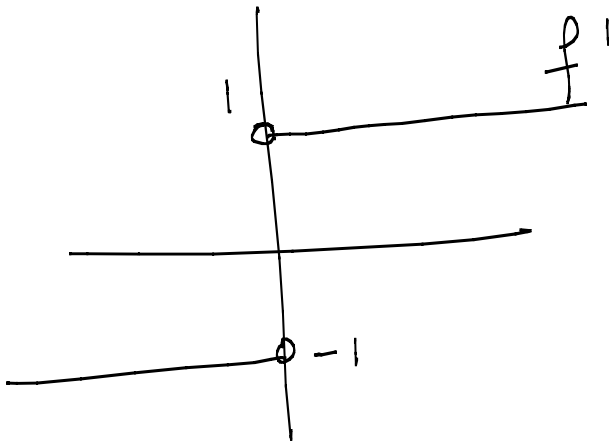
1.



- Where is f increasing? $(-\infty, -1) \cup (1, \infty)$.
- Where is f decreasing? $(-1, 1)$.
- Are there local maxima / local minima.

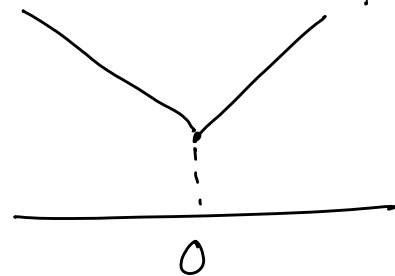
Loc max at $x = -1$
 Loc min at $x = 1$.

2.

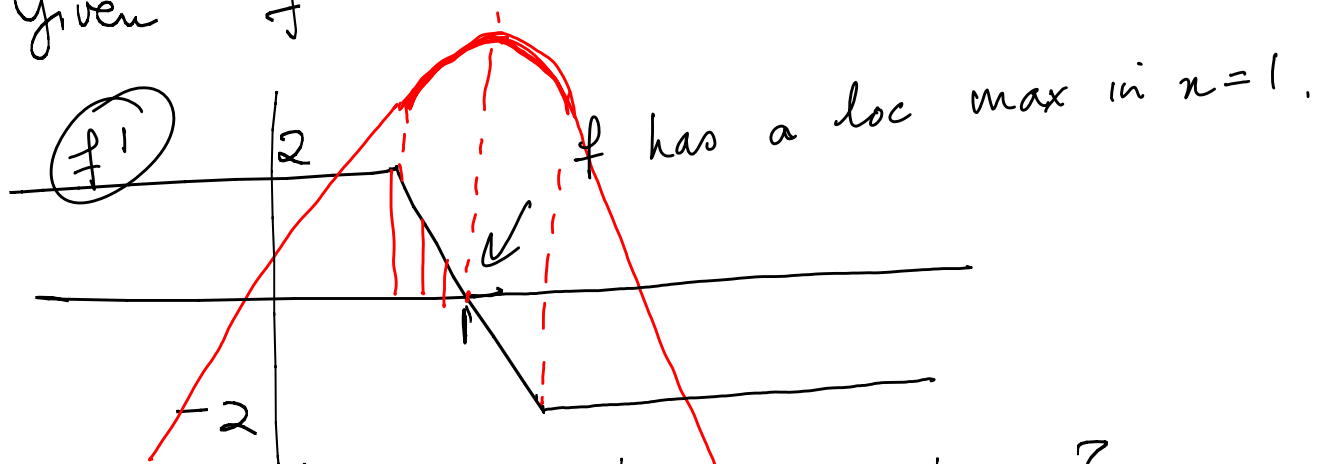


Where is f increasing?
 $(0, \infty)$

Where is f decreasing?
 $(-\infty, 0)$



Eg. Given f'



1. Where does f have a local max/min?
2. Sketch the shape of the function f .

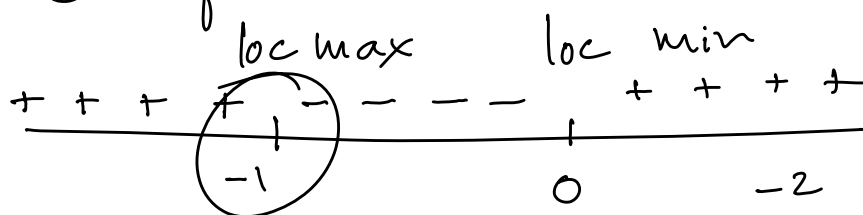
Example Determine where the function $f(x) = x^2 e^{2x}$ is increasing. Identify local extremes.

Sol

$$f'(x) = 2x \cdot e^{2x} + x^2 \cdot e^{2x} \cdot 2$$

$$= 2x e^{2x} (1+x)$$

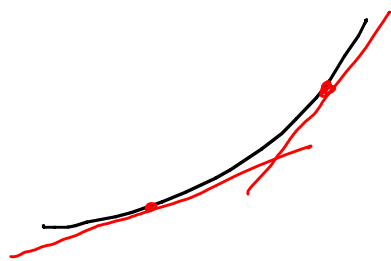
$$f'(x) = 0 \text{ if } x = 0 \text{ or } x = -1$$



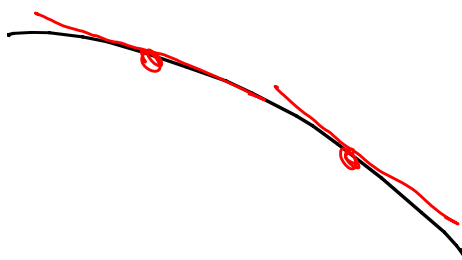
f has a loc max $f(-1) = e$

f has a loc min $f(0) = 0$.

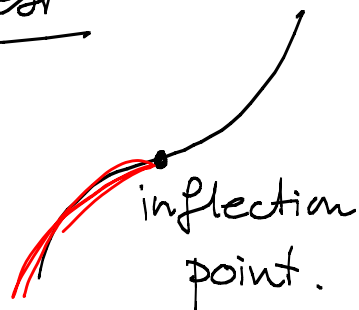
Concavity & the second derivative test



Concave up
(above its tangents)

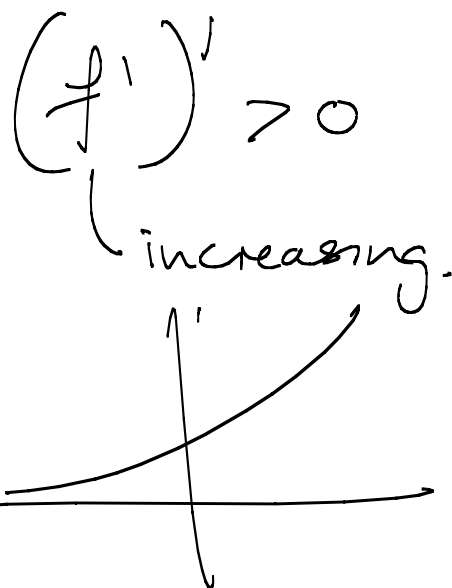


Concave down.
(below the tangents)



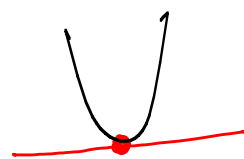
If $f'' > 0$ then f' increases
and f is concave up

If $f'' < 0$ then f' decreases
and f is concave down.

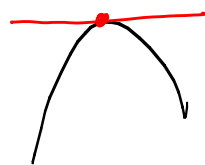


Second derivative test:

If $f'(c) = 0$ and $f''(c) > 0$ then
 f has a local minimum



If $f'(c) = 0$ and $f''(c) < 0$ then
 f has a local maximum.



Example

1. Find the local max/min of $f(x) = x^4 - 4x^2 + 4$ by using the second derivative test

$$f'(x) = 4x^3 - 8x$$

$$f''(x) = 12x^2 - 8$$

$$f'(x) = 0 \Rightarrow 4x(x^2 - 2) = 0 \\ \Rightarrow x = 0 \text{ or } x = \pm\sqrt{2}$$

$f''(-\sqrt{2}) > 0$	$f''(0) < 0$	$f''(\sqrt{2}) > 0$
$-\sqrt{2}$	0	$\sqrt{2}$
$\swarrow$	$\downarrow$	$\searrow$
loc min	loc max	loc min.

2. Find the local max/min of $f(x) = xe^x$ by means of the second derivative test.

$$f'(x) = xe^x + e^x = (x+1)e^x$$

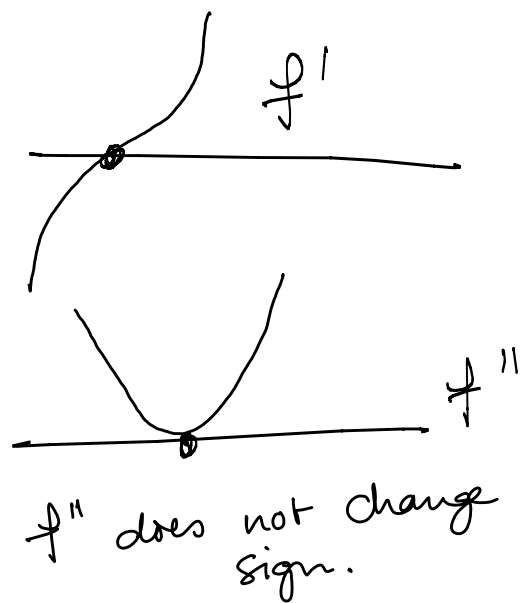
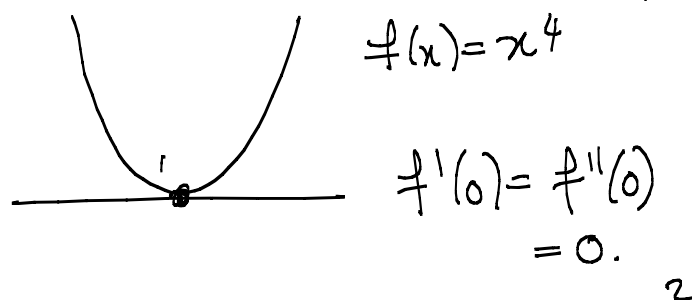
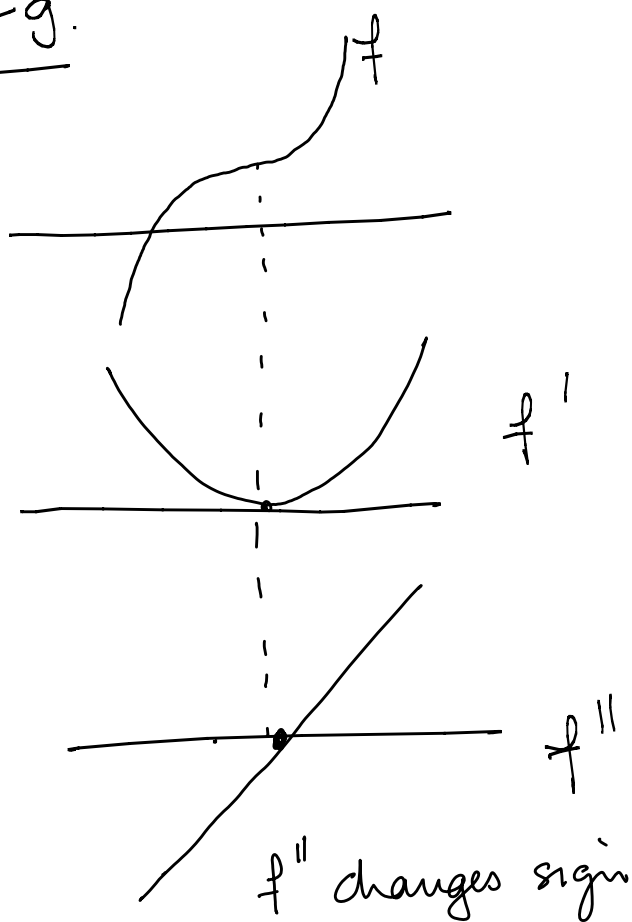
$$f''(x) = e^x + (x+1)e^x = e^x(x+2)$$

Inflection points

f has an inflection point in C if f is continuous in C and changes concavity (from up to down or from down to up)

An inflection point often occurs where $f''(c) = 0$ but that is no guarantee for an inflection point.

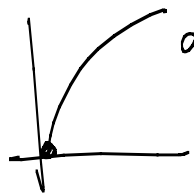
eg.



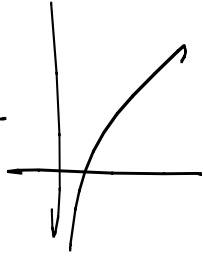
Problems on extremes etc.

1. Do the following functions have critical numbers in $x=0$?

a) $f(x) = \sqrt{x}$

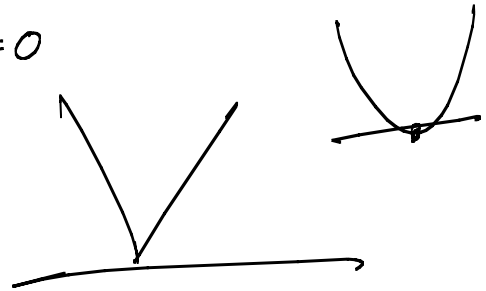


b) $f(x) = \ln x$



2. Say if the following statements are correct or not or fix them.

(1) If f has a local extreme in $x=c$ then $f'(c)=0$

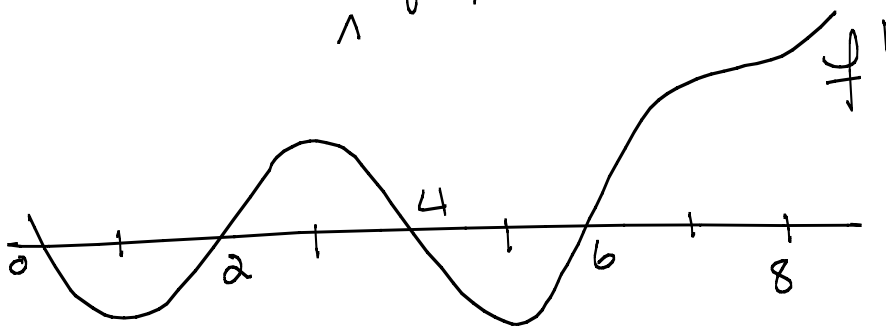


(2) If $f'(c)=0$ then f has a local extreme in $x=c$.

(3) If f has an inflection point in c then $f''(c)=0$

(4) If $f''(c)=0$ then f has an inflection point in $x=c$.

3. Consider the ^{derivative -} graph

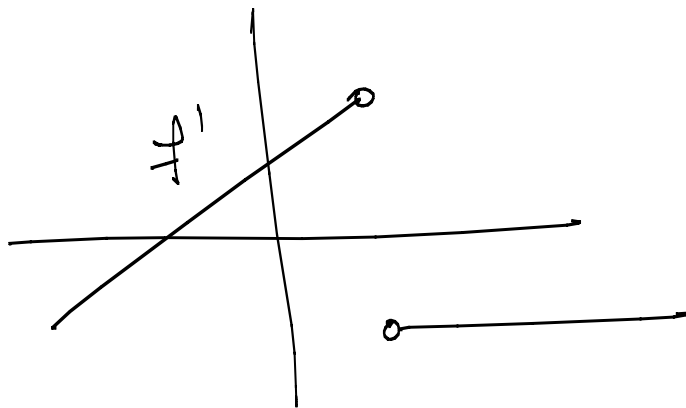


Where is f increasing?

Where does f have local maxima?

Where is f concave up?

4. f is continuous and f' looks as follows



Sketch f .

5. Consider $f(x) = \frac{\ln x}{x}$

(1.) Find the local extremes

(2) Find the absolute extremes on $[1, b]$
(Use C1M)